

An unmeasured observable in two-channel Bell experiments

Proposal for a settings-torus scan of the fourfold coincidence sum

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The observable

In a two-channel polarization Bell experiment with analyzer settings (a, b) , the fourfold coincidence sum

$$R_4(a, b) = R_{++} + R_{+-} + R_{-+} + R_{--}$$

is predicted flat by quantum mechanics under fair sampling, and has long been recognised as the natural fair-sampling diagnostic. Its angular dependence has never been bounded below the percent level: the landmark two-channel experiment reported the sum only as “typically $\sim 80 \text{ s}^{-1}$ ”, the percent-level setting-dependent anomalies in the Innsbruck raw data were attributed to pair-identification procedures rather than bounded, and the modern high-efficiency tests adopted the Eberhard/CH form — a rigorous sidestep that leaves the observable unmeasured. Full record: arXiv:2608.18886, §9.

Two results in arXiv:2608.18886 bear on this sum. First, a general one: no local model that reproduces the singlet correlation exactly through the detection channel alone can also have an angle-independent coincidence rate (Proposition 1); some angular modulation is forced. Second, a specific one: for the detection profile solved in the paper (moiré phase locking, MPL, at full detection share) the modulation is a $\cos 4(a - b)$ fringe of peak-to-peak relative amplitude 12.4%, leading harmonic exceeding its first overtone $\approx 6:1$. Hybrid models paying partly through measurement dependence show proportionally reduced amplitude; the trade-off is quantified by linear programming along that family ($\delta_f < 1\%$ forces $\geq 68\%$ of Hall’s measurement-dependence floor; $\delta_f < 0.5\%$ forces $\geq 80\%$).

Why no existing dataset bounds it. The fringe vanishes identically at the CHSH angles: $\cos(2k \cdot 45^\circ) = \cos(2k \cdot 135^\circ)$ for every harmonic k , so $R_4(22.5^\circ) = R_4(67.5^\circ)$ exactly, term by term (arXiv:2608.18886, Theorem 2). Four-point CHSH experiments — including the loophole-free tests — are mathematically blind to the observable. Only a full settings scan sees it.

The measurement

Scan the full settings torus (a, b) — e.g. 24×24 points — and Fourier-analyse $\log R_4(a, b)$. Any local apparatus response factorizes as $A(a)B(b)$ and after the logarithm is additive: it occupies only the Fourier axes, at any amplitude, harmonic content, or drift phase. The physical signal $\cos 4(a - b)$ is irreducibly joint and occupies the anti-diagonal pixels $(k, -k)$. The estimator is therefore self-vetoing against factorizable systematics. Because the fringe has known (zero) phase, the estimator is the signed projection onto the $(2, -2)$ pixel; a magnitude estimator would be biased positive under noise and is not used. The signed projection assumes the relative zero of the two analyzers is calibrated: a relative offset δ rotates signal into the quadrature and costs a factor $\cos 4\delta$ (0.3% at $\delta = 1^\circ$). The quadrature (imaginary) component of the same pixel must be null-consistent in every dataset, signal or none, and therefore serves as a built-in systematic check.

In simulation with 3% apparatus ripple and Poisson noise at 10^6 counts per point, a null input is recovered as $(0.00 \pm 0.01)\%$ and the estimator is linear from 0.2% through the full 12.4%. Poisson-limited 5σ sensitivity at $\delta_f = 0.1\%$, measured against the null distribution, is reached at $\sim 4 \times 10^5$ counts per point ($\sim 2.3 \times 10^8$ pairs); the paper quotes the deliberately conservative figure of ~ 7 hours

at 5×10^4 pairs/s, or ~ 20 minutes at 10^6 pairs/s, and that is the figure to plan against. Controls: randomized visiting order and heralded-singles normalization against source drift; interleaved reference points against slow systematics; linear-regime detector rates; quadrature null check at the signal pixel.

Requirements. A standard polarization-entangled source, independently rotatable two-channel analyzers, and coincidence rates $\gtrsim 5 \times 10^4$ pairs/s. No loophole-free rigour is needed: the observable is a rate map, not an inequality violation.

Outcomes

Every branch is publishable:

1. **Flatness at 0.1%** — the first sub-percent bound on a diagnostic left unmeasured for four decades. Through the trade-off surface of arXiv:2608.18886, it forces any local account of the singlet to carry $\gtrsim 95\%$ of Hall’s measurement-dependence floor, making measurement dependence the governing framework for local realism.
2. **A fringe of the predicted shape** — a flat-rate prediction of quantum mechanics fails.
3. **A fringe of a different shape** — the detection sector is excluded as constructed, and a joint (a, b) dependence that no factorizable apparatus model can produce is characterised.

Materials

Paper: arXiv:2608.18886 (trade-off surface, mechanism, Theorem 2, protocol §10).

Scripts, including the sensitivity simulation above (`protocol_sensitivity.py`): `Exact_Price_Source_Material` folder of the `moire-phase-space-sampler` repository (GitHub, user `RandomInternetPreson`).

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